

27th Experiment

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_classification
from sklearn.metrics import accuracy_score
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
from tensorflow.keras.optimizers import Adam

# Generate Two-Class Dataset
X, y = make_classification(
    n_samples=500,
    n_features=2,
    n_redundant=0,
    n_informative=2,
    n_clusters_per_class=1,
    class_sep=2,
    random_state=42
)

# Build Neural Network
model = Sequential([
    Dense(8, activation='relu', input_shape=(2,)),
    Dense(4, activation='relu'),
    Dense(1, activation='sigmoid')
])

# Compile Model with Learning Rate = 0.001
optimizer = Adam(learning_rate=0.001)
```

```
model.compile(  
    optimizer=optimizer,  
    loss='binary_crossentropy',  
    metrics=['accuracy']  
)
```

```
# Train Model
```

```
history = model.fit(  
    X, y,  
    epochs=50,  
    verbose=0  
)
```

```
# Training Loss Graph
```

```
plt.figure(figsize=(5,3))  
plt.plot(history.history['loss'])  
plt.title("Training Loss")  
plt.xlabel("Epoch")  
plt.ylabel("Loss")  
plt.grid(True)  
plt.show()
```

```
# Prediction
```

```
pred = model.predict(X)  
pred_class = (pred > 0.5).astype(int)
```

```
# Accuracy
```

```
acc = accuracy_score(y, pred_class)  
print("Accuracy:", round(acc*100,2), "%")
```

```
# Dataset Visualization
```

```
plt.figure(figsize=(5,4))
```

```
plt.scatter(X[:,0], X[:,1], c=y)
```

```
plt.title("Two-Class Data (ReLU, LR=0.001)")
```

```
plt.xlabel("Feature 1")
```

```
plt.ylabel("Feature 2")
```

```
plt.show()
```

Output:

